

Intermediate Code Generator Phase: (Bottom to Up)
Practical:

$a = b / (c + d) * e ;$

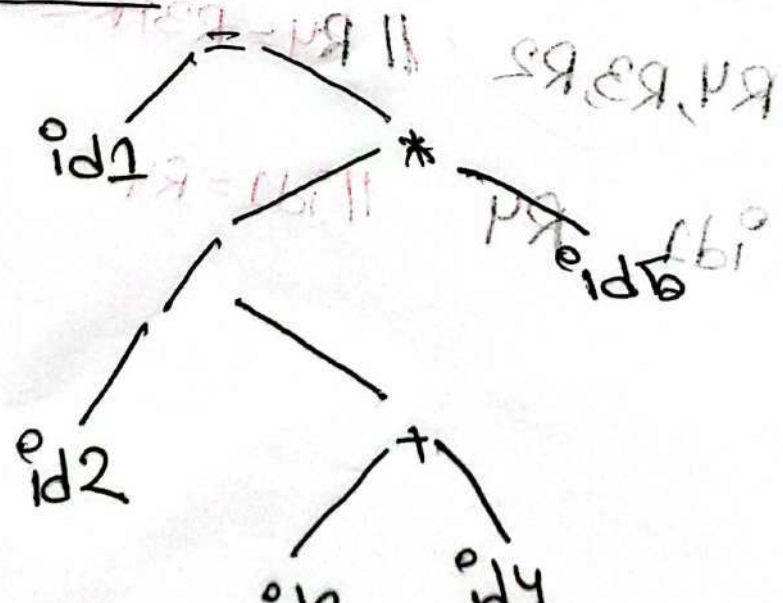
Lexical Phase:

(a = b / (c + d) * e ;)

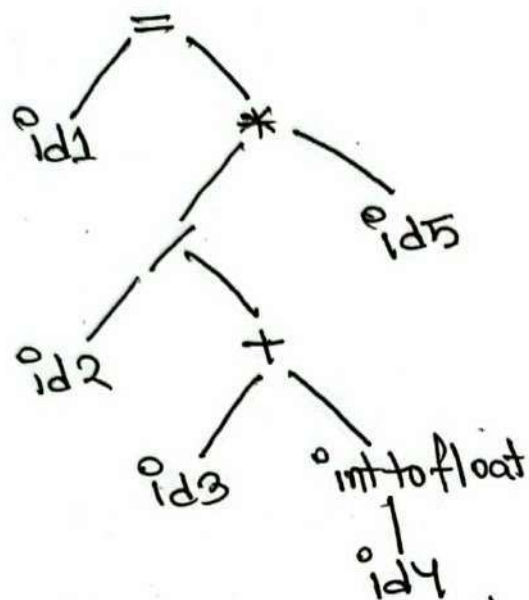
	Name	Type
1	a	float
2	b	float
3	c	float
4	d	int
5	e	float

$\langle id, 1 \rangle \langle = \rangle \langle id, 2 \rangle \langle / \rangle \langle (\rangle \langle id, 3 \rangle \langle + \rangle \langle id, 4 \rangle \langle * \rangle \langle id, 5 \rangle \langle ; \rangle$

Syntax Phases:



Semantic Phase:



Intermediate Code Generator:

`t1 = int to float(id4)`

`t2 = id3 + t1`

`t3 = id2 / t2`

`t4 = t3 * id5`

`id1 = t4`

Code Optimizer:

~~`t1 = int to float(id4)`~~

`t2 = id3 + id4` // assuming `id4` is float now.

`t3 = id2 / t2`

`id1 = t3 * id5`

Code Generator:

`LDF R1, id3`

`LDF R2, id4`

`ADDF R3, R1, R2` // `R3 = R1 + R2`

`LDF R4, id2`

`DIVF R5, R4, R3`

`LDF R6, id5`

~~`MULF R5`~~

`MULF R7, R5, R6`

`STF id1, R7`